9.4 Experimental Design

Algebra II 9.4 · Student sheet

Name _____ Period ____ Date _____

I can…

- describe a randomized comparative experiment and identify the treatment and control groups.
- tell an experiment from an observational study.
- identify problems in an experiment's design.

Vocabulary · fill in the missing word

Controlled experiment — compares a _____ group with a control group

Treatment group — the group that _____ the treatment

Control group — the group that does not (it may get a _____)

Placebo — a harmless, _____ treatment that looks like the real one

Randomized comparative experiment — subjects are randomly assigned to the _____ or control group

Replication — repeating an _____ with many subjects so results aren't due to chance

Observational study — observes subjects without _____ a treatment; it can't show cause and effect

Part 1 · Design the test

Question: **Does chewing gum improve test scores?** Plan an experiment.

Part of the plan	Your plan
Treatment group	
Control group	
How you assign students	
What you measure	
One thing that could go wrong	

Part 2 · Notes and examples (fill in each note, then work the example)

A good experiment: A good experiment has a _____ group and a _____ group, _____ assignment, and enough subjects (_____).

Example 1

A company tests a new energy drink. 100 volunteers choose whether to drink it or water before a run, and their times are compared.

Is this a randomized comparative experiment? Explain.

Example 2

60 patients are randomly assigned to take a new sleep pill or a placebo, and their hours of sleep are compared. Identify the treatment group and the control group.

Cause and effect: Only an _____ can show cause and effect.

Example 3

Tell whether each is an experiment or an observational study.

a. Researchers compare the test scores of students who eat breakfast with those who don't.

b. A teacher randomly assigns half of a class to study with flashcards.

Example 4

An experiment tests whether music improves typing speed. 10 students who like music type while listening to it, and 10 students who don't like music type in silence. Describe the problems.

Part 3 · Partner practice

Partner 1

A study records how many hours teens sleep and their grades. Is it an experiment or an observational study? Can it show that sleep causes better grades?

Partner 2

To test a new acne cream, a company gives it to 15 volunteers, and 10 of them improve. Describe the problems with the design.